

# TYLER HENDERSON

Fenestration Technical Lead | NFRC Compliance & Standards Strategy

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## PROFESSIONAL SUMMARY

Fenestration technical leader specializing in NFRC/CSA A440.2 compliance, thermal simulation, and certification strategy. ISO/IEC 17065 QMS programs, EPD/LCA service lines, and automation that cuts simulation project turnaround from weeks to days. Active in NFRC's standards process as an ANS Standards Committee member and Task Group participant.

## CORE COMPETENCIES

- NFRC / CSA A440.2 Simulation & Certification Strategy
- Compliance Program & ISO/IEC 17065 QMS Development
- EPD / LCA Program Development & Delivery
- Simulation Workflow Automation & Throughput Optimization
- Standards Development & Technical Committee Participation

## TECHNICAL SKILLS

**Engineering Simulation:** LBNL THERM & WINDOW (Advanced), Flixo, Physibel, NFRC/CSA A440.2 & Passive House methodologies  
**Compliance & Sustainability:** ISO/IEC 17065 QMS, OneClick LCA (EPD generation)  
**Automation & Data:** Python, SQL/ODBC, proprietary binary file parsing

## PROFESSIONAL DEVELOPMENT & STANDARDS INVOLVEMENT

- Member, NFRC ANS Standards Committee (2026-Present)
- Presenter, NFRC Simulators Task Group (March 2026). Presented independent technical research on two-box spacer modeling methodology and validation

## TECHNICAL ACHIEVEMENTS & PUBLICATIONS

**Validation of Two-Box Spacer Modeling Across Spacer Widths for Fenestration Thermal Analysis - Independent Technical Report (February 2026)**

- Authored an independent technical report evaluating two two-box spacer calibration strategies, per-width calibration and single nominal-width calibration, against detailed spacer models using LBNL THERM and NFRC 100/200/500 methodology.
- Ran 900 full-frame simulations across 3 representative spacer families (composite single seal, metal single seal, metal double seal) and 25 discrete spacer widths from 8-20 mm.
- Found that width-calibrated two-box models reproduce detailed spacer performance with high fidelity, holding full-window U-factor deviations to within approximately 0.008 W/m<sup>2</sup>·K across the full width range.
- Demonstrated that models calibrated at a single nominal width are not generally transferable across spacer widths without recalibration, particularly for spacers with moderate-to-high effective thermal conductivity.

## PROFESSIONAL EXPERIENCE

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### Technical Solutions Lead (R&D & Strategy) | Layton Consulting Ltd., Surrey, BC

January 2024 – December 2025

- Compliance & QMS: Engaged the Standards Council of Canada (SCC) to initiate ISO/IEC 17065 accreditation; performed gap analysis against existing organizational practices, designed and implemented QMS structure, and issued recommendations for remaining accreditation requirements, positioning the firm for expanded certification-body eligibility.
- EPD/LCA Program: Launched a new EPD service line from the ground up and led a multi-product-line engagement using OneClick LCA, delivering 6 EPDs for a major Ontario-based aluminum window manufacturer, opening a new revenue line for the firm.
- Simulation Database Platform: Architected a web-based simulation database (Next.js/Supabase) cataloguing all historical project simulations, with automatic integration into calculation spreadsheets and reporting workflows.
- Simulation SOPs: Defined and implemented Standard Operating Procedures for NFRC-compliant simulation methodology on complex glazing products, ensuring full traceability across projects.
- Workflow Automation: Built Python-based automation tools that cut project turnaround by an order of magnitude, shrinking multi-week simulation engagements to days, and multi-day engagements to hours, unlocking high-volume certification revenue previously inaccessible to the firm.
- Strategic Feasibility: Led technical feasibility analysis for long-term initiatives, including LCA/EPD implementation and data governance/privacy risk.

### Fabrication Process Specialist & CNC Lead | Glastech Glazing Contractors Ltd., Surrey, BC

August 2023 - January 2024

- Programmed and optimized RhinoFab1100 and Emmegisoft 3-axis CNC workflows for precision curtain wall fabrication.
- Redesigned warehouse layout and production flow to minimize material handling and increase throughput.
- Developed training protocols for staff on CNC operation and automated fabrication logic.

## TECHNICAL PORTFOLIO

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- NFRC-101-Materials ([github.com/hendersontylerjohn/NFRC-101-Materials](https://github.com/hendersontylerjohn/NFRC-101-Materials)) - Automated pipeline that monitors, parses, and validates NFRC 101 material property tables (Appendices A, B, C) from the published PDF, syncing them into structured, versioned JSON via a scheduled GitHub Actions workflow.
- Thm-Reader ([github.com/hendersontylerjohn/Thm-Reader](https://github.com/hendersontylerjohn/Thm-Reader)) - Rust library for reading and writing THERM 7.x binary .thm files, exposing typed access to header, polygon, and IGU/glazing data plus the material, boundary-condition, and gas libraries, with round-trip byte-accurate serialization and JSON export.

## EDUCATION & CERTIFICATIONS

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- NFRC LEAFF Certified Simulator - National Fenestration Rating Council (2026)
- Phius Blue Path Certified Simulator - Passive House Institute US (Phius)
- University of British Columbia Okanagan - B.A.Sc, Electrical Engineering (incomplete), September 2021 - May 2023; coursework in electromagnetism & optics, circuit design, embedded systems, and SolidWorks modeling.